Cut Sheet

Board Design Summary

Style: edge-grain

Dimensions: 300 mm (11.81") × 450 mm (17.72") × 38 mm (1.50")

Species: Walnut, Maple

Juice Groove: Perimeter loop, 10.0 mm (0.39") wide × 8.0 mm (0.31") deep, 20.0 mm (0.79") from the edge, corner radius 25 mm (1.00"), top face.

Rubber Feet: 4 rubber feet, 25 mm (1.00") diameter × 9.5 mm (0.38") high, foot centres 25 mm (1.00") in from the edges (underside).

Handles: handle

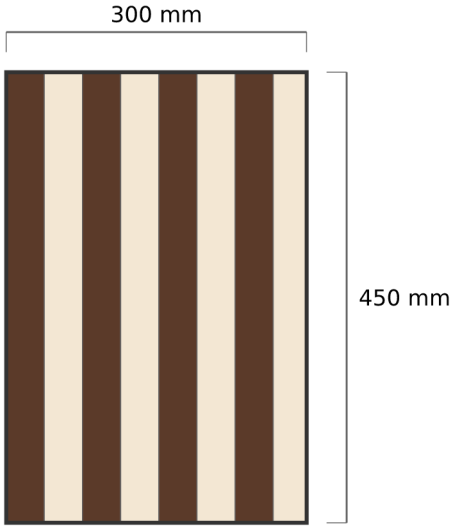
Cut List

Part	Width	Thickness	Length	Species
Board strips	25.4 mm	38 mm	450 mm	Walnut

Cut List

Part	Qty	Width	Thickness	Length	Species
Panel A: Walnut/Maple strip-set panel 300 mm (11.81") wide: 4 × Walnut 38 mm (1.50") (rip 40 mm (1.56"), then joint) + 3 × Maple 38 mm (1.50") (rip 40 mm (1.56"), then joint) + 1 × Maple 33 mm (1.31") (rip 35 mm (1.37"), then joint); too wide for the planer at 300 mm (11.81"): flatten on the router sled or drum sander; Crosscut to final length: 450 mm (17.72"); Too wide for the planer (299 mm (11.77")): flatten both faces on the router sled or drum sander	1	300 mm (11.81")	40 mm (1.56")	503 mm (19.81")	Walnut/Maple

Board Layout (Top View)



Step 1: Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 3.0 mm (0.12"). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Settings:

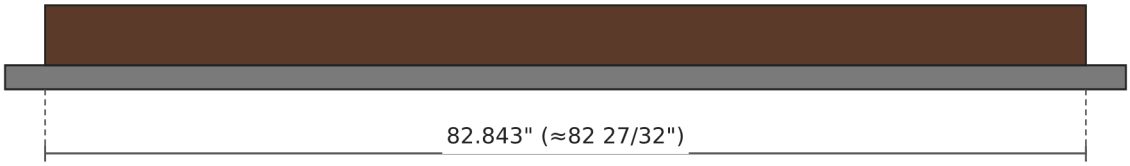
Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 305 mm (12.00") or thinner than 12.7 mm (0.50").

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 305 mm (12.00") long.

Edge Grain: step 1: Mill rough stock

Whole blank on the jointer (schematic, not to scale)



Mill and rip whole blanks before any crosscut. Blank length = max(305 mm, slices x (slice + kerf) + 2 x 25 mm trim). Never send a crosscut slice or piece shorter than 305 mm to the jointer or planer.

Walnut strip: blank 82.843" (≈82 27/32") long, 4 slices of 20.094" (≈20 3/32") crosscut after milling.

Maple strip: blank 82.843" (≈82 27/32") long, 4 slices of 20.094" (≈20 3/32") crosscut after milling.

Blank lengths: catalog blankRule applied to the app cut list by boardviz; the app export does not apply the rule yet.

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.5 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Step 2: Rip strips

Rip strips to the board's final thickness plus 3.0 mm (0.12") with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

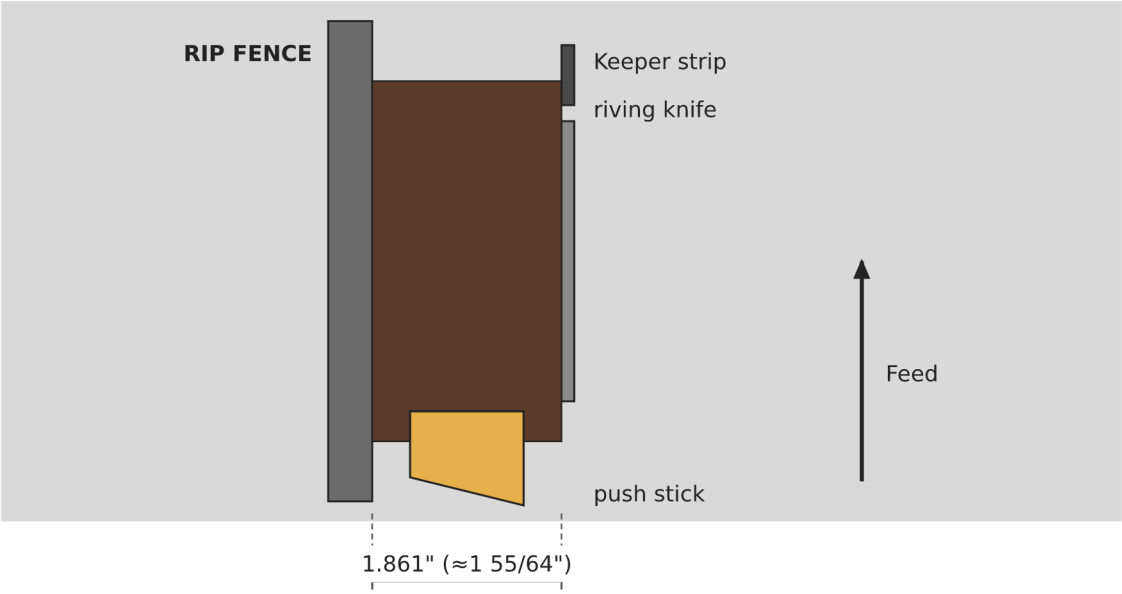
Settings:

Fence to blade: 40 mm (1.56") and 35 mm (1.37")

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 152 mm (6.00"). Never reach over or behind the blade.

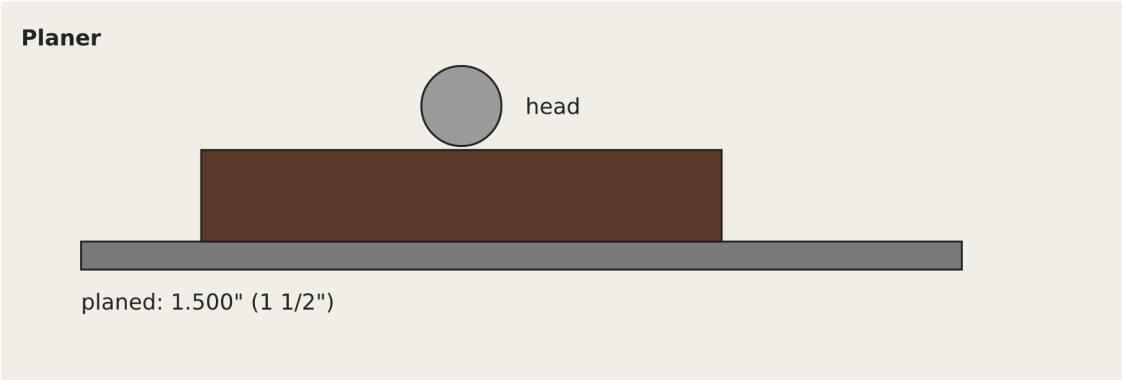
Edge Grain: rip-fence setup, step 2: Rip strips



Measure fence face → near side of a tooth

set toward the fence

Kerf: 1/8" (3.2 mm) full-kerf blade



Rip strips to the board's final thickness plus 1/8" (3 mm) with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

Fence to blade: strip width from cut list

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

Step 3: Rotate strips

Turn every strip 90° so the edge faces up. Keep growth rings alternating.

Edge Grain: strip sequence



1. Walnut
2. Maple
3. Walnut
4. Maple
5. Walnut
6. Maple
7. Walnut
8. Maple

Turn every strip 90° so the edge faces up. Keep growth rings alternating.

Step 4: Glue up

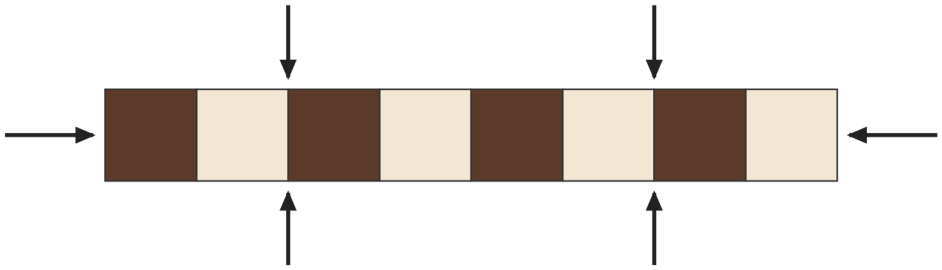
Glue strips edge to edge. Alternate clamps above and below the panel and check for flat with a straightedge.
Flatten Panel A, 300 mm (11.81") wide:
too wide for the planer at 300 mm (11.81"): router sled or drum sander.

Settings:

Clamp spacing: about 203 mm (8.00")

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Edge Grain: glue-up 1, step 4: Glue up



Glue strips edge to edge. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

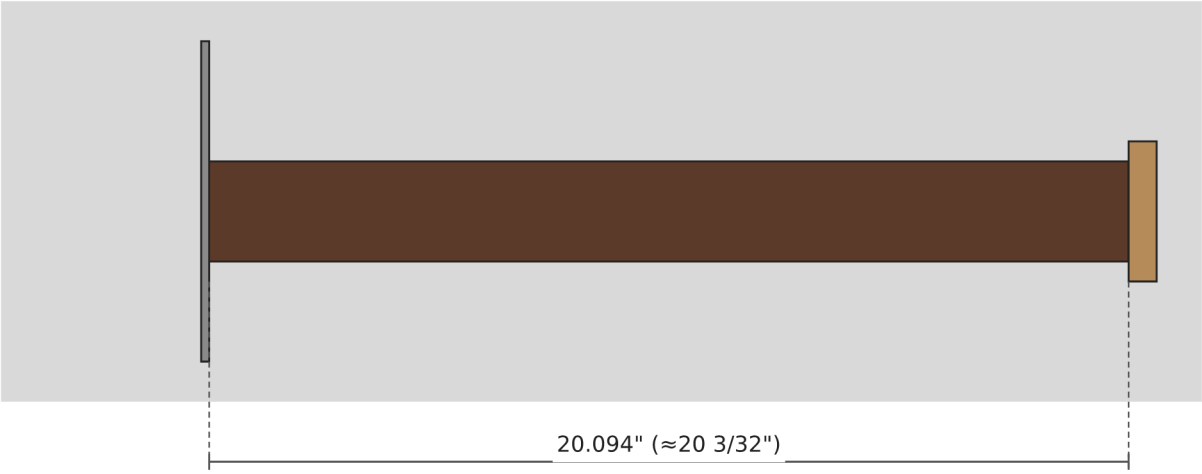
NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Step 5: Trim to length

Square both ends on a crosscut sled.
Sled width: sled must clear the glue-up, 300 mm (11.81").
Set a stop block for the final length, 450 mm (17.72"), and crosscut on the sled; never use the rip fence as a stop.

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

Edge Grain: crosscut, step 5: Trim to length



Square both ends on a crosscut sled.

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

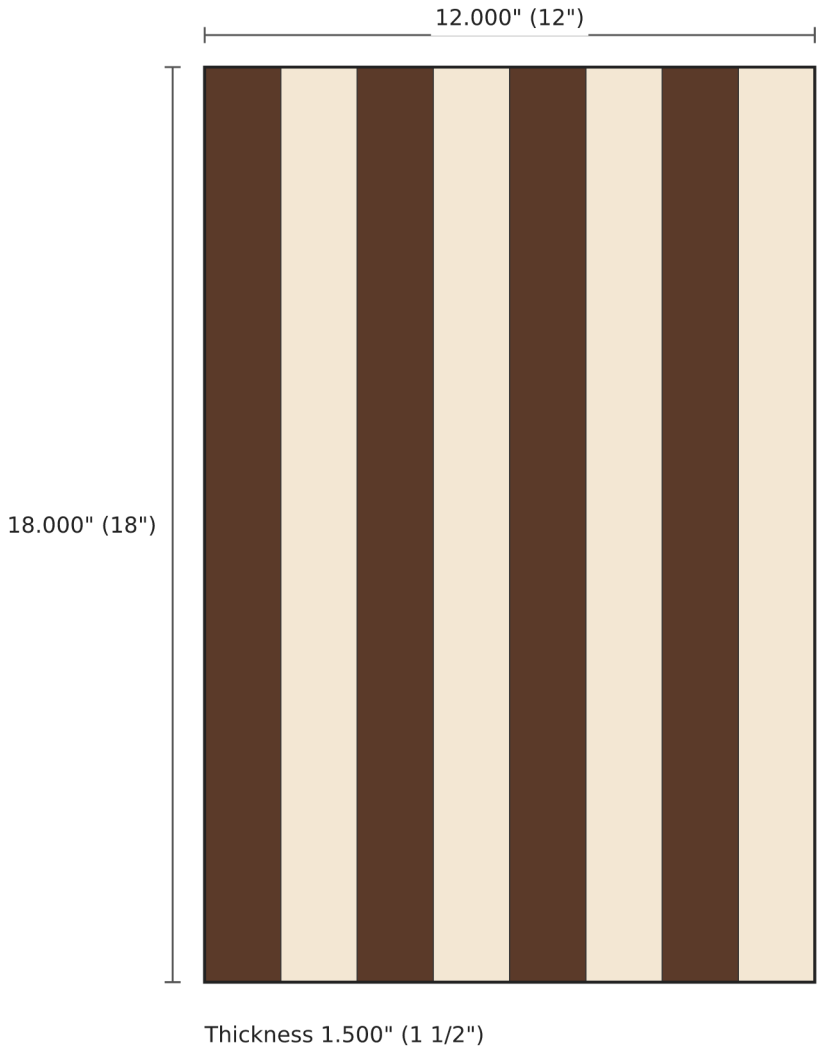
Step 6: Flatten and sand

Flatten the final glue-up, 300 mm (11.81") wide:
too wide for the planer at 300 mm (11.81"): router sled or drum sander.
Then sand: 80, 120, 180 grit. Break edges (180 grit).

Settings:

Grits: 80 / 120 / 180

Edge Grain: top view



Scrape glue, then plane flat if the panel is no wider than 12" (305 mm), otherwise use the router sled or drum sander. Sand 80, 120, 180 grit. Break edges (180 grit).

Grits: 80 / 120 / 180

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Step 7: Edge profile

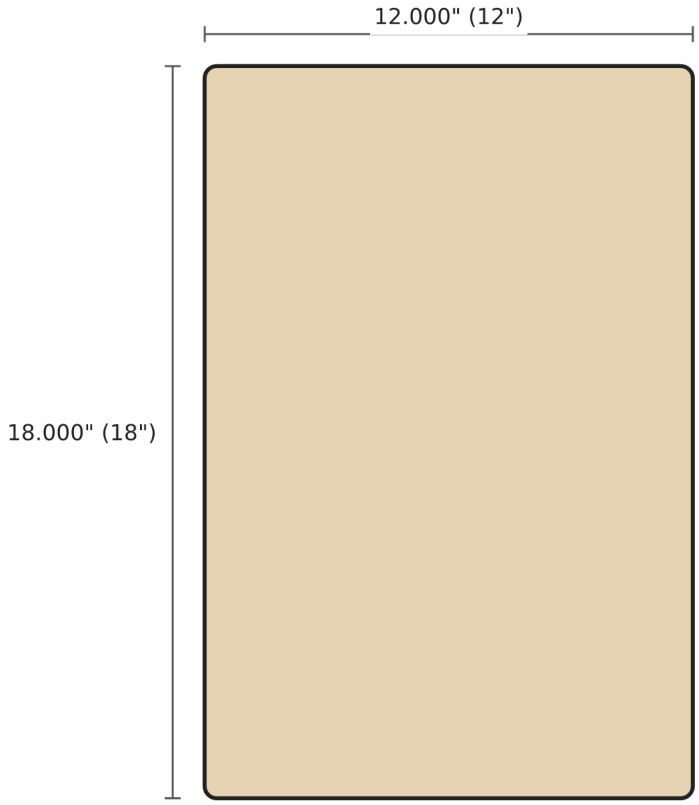
Rout the edge profile, default: 3.2 mm (0.13") roundover. Optional finger grips: core box / round-nose bit 3/8-1/2" (9.5-12.7 mm).

Settings:

Bit: 3.2 mm (0.13") roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Edge Grain: outline, step 7: Edge profile



Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8–1/2" (9.5–12.7 mm).

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Step 8: Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.